

CSE 0611-1101
Computer Fundamentals, Course outline (OBE based Curriculum)

Part A

1. Course Code : CSE 0611 1101
2. Course Title : Computer Fundamentals
3. Course Type : Core Course
4. Level : Honors
5. Academic Session : Summer 2024
6. Course Instructor : Md. Rahat Khan
7. Pre-requisite : None
8. Credit Value : 3.00
9. Contact Hours : 03
10. Total Marks : 100

11. Rationale of the course:

As digitization is touching our lives in almost all spheres computer is omnipresent due to this. Computers are present everywhere from schools to colleges to universities to professional world. Many of our daily tasks are done on the computer too. Computer Fundamentals is one prominent core course that is concerned of the basics of using a computer. This course has been designed for the newly admitted students with little to no computer experience.

12. Course Objectives:

The main objective of this course is to introduce the fundamentals of computing devices and reinforce computer vocabulary, particularly with respect to personal use of computer hardware and software, the internet, networking and mobile computing. It focuses on such computer literacy that prepares students for life-long learning of computer concepts and skills. Students discovers why computers are essential components in education, business and society in this course.

Objectives:

- To understand basics of computer and working with OS
- To develop working skills with productivity tools, graphics designing and Internet
- To acquire basic programming skills
- To apply computing in problem solving

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs):

Program Learning Outcomes (PLOs):

PLO	Area	Program Outcomes
PLO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals to the solution of complex problems of computer and information technology.
PLO2	Problem analysis	Identify, formulate, research and analyze complex Computer Engineering problems.
PLO 3	Design and Development	Design solutions for complex computer engineering problems and design system components or processes that meet the specified needs with technological concerns.
PLO4	Investigation	Conduct investigations of complex computer science and technology problems, considering experimental design, data analysis and interpretation.
PLO5	Modern tool usage	Create, select and apply appropriate techniques, resources and modern computer engineering and ICT tools.
PLO6	Engineer and Society	The engineer and society: Apply reasoning informed by contextual knowledge to develop the computer technologies for society benefits.
PLO7	Environment and sustainability	Understand the impact of professional computer Engineering solutions in societal and environmental contexts and demonstrate the knowledge for

		sustainable development.
PLO8	Ethics	Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PLO9	Individual work and teamwork	Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PLO10	Communication	Communicate effectively about complex computer science and Engineering activities with the engineering community and with society at large in both oral and written.
PLO11	Project management and finance	Demonstrate knowledge and understanding of computer engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PLO12	Life-long learning	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Course Learning Outcomes (CLOs)

After completion of the course, student should be able to-

CLO1	Develop a vocabulary of key terms related to computer systems and to software program menus.	PLO 1
CLO2	Critique the influence of computing and the internet on society including privacy, legal and ethical issues.	PLO 6, 7, 8
CLO3	Apply his/her knowledge to detect, analysis and recommended solution for the problem related to computer systems.	PLO 2, 3

Mapping CLO with BNQF:

No. of CLOs	BNQF Domain	Blooms Domain	PLO mapping
CLO1	F ₁	Understand	PLO 1
CLO2	S ₆ , P ₄	Evaluate	PLO 6, 7, 8
CLO3	F ₃	Apply	PLO 2, 3

Marks Distribution on CLOs:

CLOs	Continuous Assessment (30)			Mid term Exam (30)		Final Exam (40)		Total (100)
	Class Performance (10)	Class Test (10)	Assignment (10)	MCQ (10)	SAQ (20)	MCQ (10)	SAQ (30)	
CLO-1	3	5	N/A	5	10	2	10	35
CLO-2	3	5	5	N/A	N/A	4	10	27
CLO-3	4	N/A	5	5	10	4	10	38

Topics to be covered/ content of the course-

Lesson	Topic	Teaching Learning Strategy	Assessment Strategy	Corresponding CLOs
1.	Introduction and Motivation - Course outline discussion - Real life application discussion Specific outcomes: - Demonstrating how course concepts apply to real-world situations increases student interest and motivation to learn. - Students gain insight into how theoretical			CLO 1

	knowledge can be utilized in practical settings, preparing them for future professional challenges.			
2.	Basic operation using PC, hardware and peripheral introduction - Demo of computer motherboard explanation Specific outcomes: - To identify and understand the function of crucial motherboard components such as the CPU, RAM slots, and expansion ports. - By recognizing how components interact on the motherboard, students can better diagnose and solve hardware issues.			CLO 1
3.	Computer generations and history. Specific outcomes: - Students gain an understanding of the progression from vacuum tubes to transistors, integrated circuits, and microprocessors, highlighting key innovations in each generation. - By studying the history of computer generations, students can appreciate how past advancements have shaped current technologies and influence future developments.			CLO 1
4.	Basic computer organization - Computer working principle Specific outcomes: - Learn how a computer processes instructions through the sequential steps of fetching, decoding, and executing, which is fundamental to how computers operate.			CLO1
5.	Computer Hardware's - I/O devices & its brief Specific outcomes: - Understanding devices like keyboards and mice for entering data into the computer. - Learning about monitors and printers for displaying information from the computer.			CLO1
6.	Computer Memories (Cache, RAM, ROM, SSD, HDD...) Specific outcomes: - Understanding the differences in speed and function between cache, RAM, ROM, SSD, and HDD. - Learning the specific roles of each memory type in data storage and retrieval.			CLO1
7.	Computer software's - Operating Systems Specific outcomes: - Understanding how operating systems manage hardware resources and provide a			CLO1

	platform for running applications. - Learning how users interact with the OS through interfaces and commands for tasks like file management and software execution.			
8.	Number system and Conversions - Computer Arithmetic Specific outcomes: - Exploring methods to convert between binary, decimal, and hexadecimal numbers. - Understanding how computers perform addition, subtraction, multiplication, and division on binary numbers.			CLO3
9.	Boolean Algebra Specific outcomes: - Understanding basic operations like AND, OR, and NOT gates. - Learning how Boolean algebra principles are applied in designing digital circuits and logic gates.			CLO3
10.	Logic gates Specific outcomes: - Understanding basic logic gates such as AND, OR, NOT, NAND, NOR, XOR, and XNOR. - Learning how these gates process binary inputs to produce specific outputs, crucial for digital logic design and computer circuitry. -			CLO3
11.	Class Test 1 & Questions Analysis	Q & A	Class test will be taken.	CLO3
12.	Working with productivity package Excel, PowerPoint. Specific outcomes: - Learning to use formulas and functions for calculations and data analysis. - Creating effective presentations with slides, graphics, and animations.			CLO 1
13.	Using Google tools for education (Google Classroom, Calendar, Gmail) Specific outcomes: - Organizing assignments, discussions, and materials for students. - Syncing assignments and due dates from Google Classroom.			CLO 1
14.	Mid-term Overview.			

15.	Networking Fundamentals - Transmission Mediums Specific outcomes: - Understanding wired (e.g., Ethernet, fiber optics) and wireless (e.g., Wi-Fi, Bluetooth) transmission mediums. - Learning how different mediums are selected based on factors like speed, distance, and environment in network setups.			CLO2
16.	Network Topologies Specific outcomes: - Examining common network topologies such as star, bus, ring, and mesh. - Applying each topology's configuration and assessing benefits like scalability and fault tolerance.			CLO2
17.	Internet, its features, usages, impacts. Specific outcomes: - Understanding the global network's reach and accessibility. - Utilizing platforms for information dissemination and collaboration.			CLO2
18.	Network Securities (Virus, Worms, Hacking... etc) Specific outcomes: - Addressing vulnerabilities and implementing safeguards against viruses, worms, and hacking. - Safeguarding information integrity and confidentiality from cyber threats.			CLO2
19.	Intellectual property - ICT Act / Digital Security Acts. Specific outcomes: - Implementing measures to secure digital assets and comply with legal frameworks.			CLO2
20.	E-commerce, E-payments Specific outcomes: - Ensuring safe and efficient electronic transactions through various payment gateways and platforms.			CLO2
21.	Computer programming history & Basics Specific outcomes: - Tracing the development from early machine language to high-level programming languages. - Understanding basic programming constructs like variables, loops, conditionals, and functions.			CLO3

22.	Develop logical concept of problem solving - Flowchart/Algorithms Specific outcomes: - Visualizing problem-solving steps using flowcharts to outline processes and decisions.			CLO3
23.	Basic program structures and variables - Data types, variables, constant Specific outcomes: - Structuring systematic procedures (algorithms) for solving problems efficiently, ensuring clear logic and steps.			CLO3
24.	Class Test 2 and Question analysis	Q & A	Class test will be taken	CLO2
25.	Basic problem solving (Practice) Specific outcomes: - Practicing coding exercises to apply algorithms in programming languages for real-world applications.			CLO3
26.	Discussion about advanced tools (Version control, Paraphrasing, Research related etc...) Specific outcomes: - Facilitating teamwork through efficient code management and version tracking. - Ensuring integrity in academic and professional writing by using tools to create unique content.			CLO3
27.	Working with productivity package (Word, LaTeX) Specific outcomes: - Creating and formatting documents with text, images, and tables for various purposes. - Writing equations and mathematical symbols with ease using LaTeX's built-in capabilities.			CLO3
28.	Final term exam overviews.			

Text Books and Reference Books:

1. Computer Fundamentals by Pradeep K. Sinha, 6th Edition.
2. Computer Fundamentals and ICT by M. LutfarRahman , M. Shamim Kaiser , M. Ariful Rahman
3. Introduction to Computer by Peter Norton.
4. Introduction to Information System by James A. O'Brien, 8th Edition.
5. Fundamentals of Computers by V. Rajaraman and N. Adabala, 6th Edition.